

KARTHICK RAJA M

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10+
YEARS ENGINEERING

20,230+
DOWNLOADS · AICHARGEWORKS.COM

96.2%
ALL-PASS RATE · EFA PAPER

70%
PRODUCTIVITY GAIN · REX @ APPIAN

01 / SUMMARY

Highly accomplished, results-oriented Lead AI/ML Engineer with over a decade of expertise leading the design and development of enterprise-scale intelligent systems across Python, Microsoft technologies, and cloud platforms (Azure, AWS). Proven ability to leverage advanced ML frameworks and Large Language Models to solve complex business problems and drive innovation. Specializes in Agentic AI, Multi-Agent Systems, RAG-based retrieval, LLM fine-tuning & training, LLM observability & tracing, LLM-as-Judge evaluation pipelines, and multi-tenant AI platform architecture.

02 / AI-ML EXPERTISE

AI-powered, data-driven applications using Python, FastAPI, Pandas, NumPy, SciPy, Scikit-learn, PyTorch, OpenAI, Gemini, DeepSeek, Llama, AWS Bedrock, LangChain, LangGraph, FAISS, DuckDB, Redis, and Streamlit.

Custom LLM training & architecture — built **TinyStories-24.5M**, my own 24.5M-parameter decoder-only LLM trained from scratch (Llama 2 architecture: RoPE, SwiGLU, RMSNorm, Flash Attention; perplexity 8.65, published on HuggingFace with live inference), plus models up to 100M parameters with custom tokenizers. Prompt engineering, tool use / function calling, AI-driven analytics, LLM observability & tracing (OTel, OpenInference), LLM-as-Judge evaluation pipelines, multi-tenant AI platform architectures.

LoRA fine-tuned a 9B-parameter model to **91.7% domain accuracy** on a custom evaluation suite (bf16 LoRA, single-GPU RTX 5090).

Designed and shipped multiple **MCP servers** — project-memory, messaging, licensing-admin, and ecosystem control.

Built a production **voice AI agent**: Alexa skill → FastAPI relay → LLM intent routing → 20 action types, with AWS request-signature verification and 98% test pass rate.

03 / OPEN SOURCE — 6 PyPI PACKAGES · 15,400+ DOWNLOADS

- 4.3k

tracemaid — execution tracing, call graphs, performance profiling.
- 4.1k

rubricon — spec-first LLM generation; EFA paper reference implementation.
- 2.6k

distill-json — JSON compression for LLMs; 60–85% token reduction.
- 2.2k

semantic-llm-cache — semantic caching; 20–40% LLM cost reduction.
- 1.4k

clinotes — git-native MCP project memory for AI coding agents.
- 0.9k

langgraph-crosschain — cross-chain node communication for LangGraph.

20,230+ total downloads across the aichargeworks.com ecosystem — incl. **TinyStories-24.5M** (own LLM — huggingface.co/karthyick) · 8 VS Code extensions with **4,700+ installs** · novel classifiers (Outside-In, MagneticClassifier) · live tools (Mermaid2Arch, Transformer Visualizer, Quantum Gates Explorer).

04 / RESEARCH PUBLICATION

“Evaluation-First Generation: Specification-Driven LLM Output Quality via Dynamic Rubric Conditioning and Iterative Criteria Refinement” (2026) — proposes the Evaluation-First Architecture (EFA): query-specific evaluation criteria generated **before** generation as structured conditioning targets. Introduces CMPG (Criteria-Masked Progressive Generation) and FWRL (Failure-Weighted Reattention Loop); **96.2% All-Pass Rate** on MT-Bench, outperforming seven baselines incl. Self-Refine (92.5%) and Best-of-5 (91.2%).

05 / FULL STACK & CLOUD

Modern web applications using React, Angular, Flask, FastAPI, .NET Core, EF, SQL, Oracle, Cosmos, DynamoDB, DuckDB, Web API — with Fluent Validation, unit testing, Fortify Scan, authentication, multi-tenant and application security. Specialized in Azure, AWS, and OpenShift for IaaS, PaaS, and hybrid cloud (full service lists in Tech Stack).

// TECH STACK

AI / ML

Machine Learning · Deep Learning · Neural Networks · NLP · CNN · RNN · TensorFlow · Scikit-learn · LLM & Gen AI · RAG · AI Agents · Multi-Agent Systems · Agentic AI · Custom LLM · LoRA · Transformer · LLM Fine-tuning · Model Tools & Chains · LLM-as-Judge · Online & Offline Evaluation · LLM Observability · OTel · Prompt Hooks & Rules · OpenAI · Gemini · Ollama · Bedrock Claude · Hugging Face

SERVER

Python · C# · MVC · Web API · Dapper · EF Core · DI · FastAPI · Flask · Redis · UV · Pydantic · Microservices · n8n · Claude Code

CLIENT

React JS · Angular · Blazor · .NET Core · TypeScript · Flutter · Tailwind & Bootstrap · Three.js · NextJS · Streamlit

DATA & BI

MS SQL · Oracle · PostgreSQL · Databricks · Chroma · DuckDB · Alembic · SciPy · Plotly · Matplotlib · Splunk · RDS · Power BI · Dynamo · S3 · Blob

CLOUD & DEVOPS

AWS (EC2 · ECR · Bedrock · VPC · IAM · Scaling) · Azure (AD · App Service · Functions) · Azure DevOps · Docker · K8s · OpenShift · SonarQube · Jenkins · GitLab CI/CD · Argo CD · Helm · Terraform

TOOLS & IDE

Git · GitLab · TFS · Kiro · Claude Code · Cursor · MCP · n8n · Obsidian · Jira · Confluence · VS Code · Visual Studio · Jupyter · Colab · Kaggle · Hugging Face · Qiskit

CERTIFICATIONS

Quantum Computing — MIT xPRO
Gen AI — Chennai Mathematical Institute
Azure — AZ-900 · AI-900 · DP-900 · PL-900

// TIMELINE

- Appian**
Lead AI Engineer
2025 - PRESENT · CHENNAI
- DHL IT Services**
Senior Software Engineer
06/2021 - 2025 · CHENNAI
- Wells Fargo India**
Technology Specialist
06/2018 - 06/2021 · CHENNAI
- Sutherland Global Services**
Senior Software Engineer
02/2017 - 06/2018 · CHENNAI
- SLK Software Services**
Software Engineer
2015 - 02/2017 · BANGALORE

// EDUCATION

- Quantum Computing Introduction**
Massachusetts Institute of Technology
LATEST
- Postgraduate, AI & Machine Learning**
IIT Roorkee
- Postgraduate, Cloud Computing**
Great Lakes Institute of Management
- Bachelor of Engineering (CSE)**
Karpagam Academy of Higher Education

// AWARDS

- Vespa Award — DHL**
Custom algorithm optimizing operational efficiency.
- Best Employee of Q — Wells Fargo**
High-impact delivery across horizontal domains.
- Best Employee — Sutherland**
Custom deployment tool improving automation.

06 / HIGHLIGHTS

Designed and proposed REX — an AI-native autonomous engineering pipeline — to Appian; adopted and implemented, delivering a ~70% productivity gain in engineering delivery.

Published 6 open-source Python packages on PyPI (tracemaids, rubricon, distill-json, semantic-llm-cache, clinotes, langgraph-crosschain) — 15,400+ downloads; 20,230+ across the full open-source ecosystem incl. 8 VS Code extensions (4,700+ installs).

Built TinyStories-24.5M — own LLM trained from scratch, published on HuggingFace — and designed enterprise-grade LLM observability platforms replacing commercial vendor dependencies.

Trained custom tokenizers and built evaluation pipelines that improved automated scoring from 25/100 to 97/100.

Architected multi-tenant AI services handling customer-sensitive data with JWT-based isolation; authored ADRs and comparative analyses that drove architectural decisions across teams.

07 / CI-CD, DEVOPS & INFRASTRUCTURE

CI/CD & DevOps: Helm Charts, Argo CD, Docker multi-stage builds, SonarQube.

Infrastructure as Code & Deployment: EKS with Helm Charts, ArgoCD GitOps, GitLab CI/CD pipelines, Docker multi-stage builds, AWS ECR registry.

08 / AGILE & ENGINEERING PRACTICES

Agile development using Jira, Confluence, GitLab (MR reviews, issue tracking), and Scrum methodologies.

Architectural Decision Records (ADRs), OpenAPI spec generation, technical design documents, multi-service API design, API desk compliance.

Pytest with coverage reports, Pydantic v2 validation, Ruff linting, type hints, pre-commit hooks, code-review practices.

Monorepo to multi-MR splitting, feature-branch workflow, cross-team collaboration for requirements-to-code bridging.

09 / INDUSTRY DOMAINS

Enterprise AI Platforms (Appian): LLM observability and evaluation infrastructure for the organization's AI vision — a low-code platform powering enterprise applications.

Logistics (DHL): Supply-chain slotting optimization algorithms, warehouse virtualization with 2D/3D transformation, predictive analytics for SKU placement.

Banking (Wells Fargo): Statement communication, governance and decommission systems, compliance-driven application development.

Insurance (Sutherland): Claims-initiation automation with RPA, SMS, and email integration.

Banking (SLK): Net-banking application development for M&T Bank.

10 / AUTONOMOUS AI DEV-PIPELINE PLATFORM — INDEPENDENT R&D

Multi-tenant SaaS architecture on enforced binary distribution (PyArmor); an autonomous dev → review → QA → E2E-UI-test multi-agent factory with multi-LLM orchestration — intelligent routing across cloud and on-premise models with automatic fallbacks.

PYTHON · FASTAPI · AZURE · COSMOS DB · POSTGRESQL · REACT · PLAYWRIGHT · MULTI-AGENT · LICENSE ENFORCEMENT

AI-PLATFORM — LLM OBSERVABILITY & EVAL

APPIAN

Custom LLM tracing and evaluation service for Appian’s Composer AI platform — end-to-end observability for multi-step AI agent workflows with enterprise multi-tenant isolation.

- Fully custom, vendor-agnostic tracing service replacing Phoenix, Datadog, and LangSmith — DynamoDB tenant partition keys for per-customer data isolation.
- LLM-as-Judge evaluation pipeline with domain-specific rubrics: automated scoring improved 25/100 → 97/100 across enterprise domains.
- Trace-correlation middleware and span processors on OTel/OpenInference standards, spanning Java monolith and Python microservice boundaries.

PYTHON · FASTAPI · BEDROCK · DYNAMODB · S3 · OTEL · OPENINFERENCE · JWT · REACT · POSTGRESQL · DOCKER · EKS · HELM · ARGO CD · GITLAB CI/CD

AI RECORDS DESIGNER — INTELLIGENT APP BUILDER

APPIAN

AI-powered dynamic records management system replicating Appian’s Records Designer — natural-language application generation with multi-agent orchestration and real-time execution transparency.

- LangGraph multi-agent orchestration: PlannerAgent coordinating Records, ProcessModel, and Interface agents through seven cross-chain workflows via custom langgraph-crosschain.
- LLM planning engine on Bedrock Claude 3.5 Sonnet replacing static keyword routing — deterministic structured outputs at temperature 0.3 with dynamic capability extraction from agent docstrings.

PYTHON · FASTAPI · BEDROCK · LANGCHAIN · LANGGRAPH · REACT · REDUX · DUCKDB · CHROMA · SSE · DOCKER

REX — SELF-LEARNING ORCHESTRATION META-LOOP

APPIAN

Self-designed autonomous learning layer over a 12-phase LLM code-generation pipeline — independently proposed to Appian (proposal.aichargeworks.com), adopted and implemented, delivering a ~70% engineering productivity gain.

- Each cycle generates EFA evaluation criteria, scores phases per-criterion, and computes a weighted hinge loss; a post-cycle meta-loop applies adaptive learning-rate scheduling with guarded dampening and atomic updates to routing/timeout/retry weight stores (W0–W3) with integrity checks and auto-restore.
- Timeout growth caps and 30-day half-life decay keep recent signal dominant; learned per-project LLM routing via a hybrid executor with trust-gated training. Plan → Execute → QA → Final QA runs overnight — tasks submitted at night ship by morning with zero human intervention.

PYTHON · LLM ROUTING · FWRL REWEIGHTING · HINGE-LOSS OPTIMIZATION · JSONL LOSS STORES

SOLVE

DHL

Slotting product for DHL supply chain management — predictive analytics generating slotting moves for SKUs, improving the slotting mechanism across DHL warehouses.

- Managed re-engineering of the slotting improvement algorithm and 2D transformation of warehouse virtualization.
- Architecture, development, cloud migration, and team coordination — client discussions converting requirements into technical tasks.

PYTHON · REACT · ASP.NET CORE · THREE.JS · C# · BLAZOR · MS SQL · AZURE · AZURE DEVOPS · CLUSTERING · SPLUNK · OPENAI · FORTIFY · DUENDE · AZURE AD · MFA · JENKINS

TRANSPORT METRICS

DHL

Medium for inbound supply-chain data from multiple vendors.

- Independently designed, developed, enhanced, and deployed as the sole contributor.

PYTHON · AZURE CORE TOOLS · AZURE FUNCTIONS · BLOB · GEN AI · PROMPT ENGINEERING

STATEMENT COMMUNICATION & URL

WELLS FARGO

Email-statement approval workflow for the Wells Fargo entity, plus a governance application deprecating and approving domains across the entity.

- Full Agile cycle — requirements gathering and UI/UX wireframes with the Product Owner.
- Innovative pilot of the URL decommission system, designed and delivered in ASP.NET Core.

ANGULAR · WEB API · C# · MS SQL · KENDO UI · PCF · SPLUNK · FORTIFY SCAN

CLAIM INITIATION

SUTHERLAND

Automated the claim-initiation process integrating SMS and email communication — processing inputs from RPA, SMS, and email for efficient claim handling with real-time notifications and reports. Full SDLC with .NET MVC; test-driven development using Visual Studio MS Unit Test.